6. QB - Array question Bank

🔰 Part 1: Basic Array Questions (1–25)

These focus on understanding syntax, creation, iteration, and simple operations.

1. Create an array of 10 integers. Initialize it and print all elements.
2. Read 5 integers from user using `Scanner` and store them in an array. Print the array.
3. Find the maximum value in an array.
4. Find the minimum value in an array.
5. Calculate the sum of all elements in an array.
6. Calculate the average of array elements.
7. Count even and odd numbers in an array.
8. Count how many positive and negative numbers are in an array.
9. Search for a given element in the array (Linear search).
10. Print elements at even indices.
11. Print elements in reverse order.
12. Copy all elements of one array into another array.
13. Merge two arrays into a third array.
14. Remove duplicates from an array.
15. Sort the array in ascending order.
16. Sort the array in descending order.
17. Swap the first and last elements of an array.
18. Rotate the array to the right by 1 position.
19. Find the second largest number in an array.
20. Find the frequency of each element in an array.
21. Count number of elements greater than a given number.
22. Find the sum of only even elements in an array.
23. Find common elements between two arrays.
24. Find the difference between the largest and smallest element in the array.
25. Replace all negative numbers in an array with 0.

🧠 Part 2: Intermediate (Logic-based) Questions (26–40)

Focus shifts to conditions, nested loops, and slightly deeper logic.

26. Read 10 marks from user and print grade for each (use `if-else`):
  - =90: A, >=80: B, >=70: C, >=60: D, else: F
27. Check if the array is sorted in ascending order.
28. Count how many times a number appears in an array.
29. Replace every element with the product of all other elements.
30. Find all pairs of elements in array whose sum is equal to a given number.
31. Shift all zeros to the end without changing the order of non-zero elements.
32. Count total number of duplicate elements in an array.
33. Separate even and odd numbers into two different arrays.
34. Create an array of strings. Print all strings starting with a vowel.
35. Input student marks and display pass/fail status for each (pass if >=40).
36. Reverse only the even numbers in the array, leave odd numbers in place.
37. Print the first repeating element in an array.
38. Move all negative numbers to the beginning and positive to end.
39. Input an array, then input a number `k`. Print all elements greater than `k`.
40. Remove all occurrences of a specific element from the array.

📦 Part 3: POJO + Array Use Cases (41–50)

These questions use custom model classes and object arrays.

Sample POJO:

```java
public class Student {
    int id;
    String name;
    int marks;

    // Constructor
    public Student(int id, String name, int marks) {
        this.id = id;
        this.name = name;
        this.marks = marks;
    }
}
```

41. Create an array of 5 `Student` objects. Initialize and print all student names.
42. Find and print the student with highest marks.
43. Print names of students who scored more than 75.
44. Count how many students passed (marks >= 40).
45. Sort the array of `Student` objects based on marks.

46. Input student data from `Scanner` and store in object array.
47. Print name of the topper (highest marks).
48. Print all students whose name starts with 'A'.
49. Check if a student with a particular name exists in the array.
50. Print the average marks of all students.

Document by ***Suyash*** 😇